```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('matches.csv')
df.head()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
0	335982	Bangalore	2008-04-18	BB McCullum	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	field	Kolkata Knight Riders	runs	140	223	20	N	NaN	Asad Rauf	RE Koertzen
1	335983	Chandigarh	2008-04-19	MEK Hussey	Punjab Cricket Association Stadium, Mohali	0	Chennai Super Kings	Rajasthan Royals	Chennai Super Kings	bat	Chennai Super Kings	runs	33	241	20	N	NaN	MR Benson	SL Shah
2	335984	Delhi	2008-04-19	SC Ganguly	Feroz Shah Kotla	0	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	Delhi Daredevils	runs	9	238	20	N	NaN	Aleem Dar	GAF Cowie
3	335985	Mumbai	2008-04-20	MV Boucher	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Royal Challengers Bangalore	runs	5	234	20	N	NaN	SD Ranade	AV Jayaprakash
4	335986	Kolkata	2008-04-20	DJ Hussey	Eden Gardens	0	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	runs	5	222	20	N	NaN	BF Bowden	K Hariharan

```
df.shape
```

```
(1095, 20)
```

```
df.info()
```

```
<class>
RangeIndex: 1095 entries, 0 to 1094
Data columns (total 20):
 id               1095 non-null   int64
 city             1044 non-null   object
 date             1095 non-null   object
 player_of_match  1090 non-null   object
 venue            1095 non-null   object
 neutral_venue    1095 non-null   int64
 team1            1095 non-null   object
 team2            1095 non-null   object
 toss_winner      1095 non-null   object
 toss_decision    1095 non-null   object
 winner           1092 non-null   object
 result           1095 non-null   object
 result_margin    1076 non-null   float
 target_runs      1095 non-null   int64
 target_overs     1095 non-null   int64
 super_over       1095 non-null   object
 method           1095 non-null   object
 umpire1          1095 non-null   object
 umpire2          1095 non-null   object
dtypes: int64, object
memory usage: 1.1 MB
```

```
df.isnull()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False	False

```
df.isnull().sum
```

```
<bound_method>
```

```
df.isnull().sum()
```

```
id                   0
city                51
date                 0
player_of_match     5
venue                0
neutral_venue        0
team1                0
team2                0
toss_winner          0
toss_decision        0
winner               3
result               0
result_margin       19
target_runs          0
target_overs         0
super_over           0
method               0
umpire1              0
umpire2              0
dtype: int64
```

```
df.describe()
```

	id	neutral_venue	result_margin	target_runs	target_overs
count	1.095e+03	1095.000000	1076.000000	1095.000000	1095.000000
mean	7.455e+05	0.000000	17.284386	166.000000	19.999000
std	3.315e+05	0.000000	21.773456	29.000000	0.000000
min	3.359e+05	0.000000	1.000000	43.000000	5.000000
25%	3.359e+05	0.000000	5.000000	154.000000	20.000000
50%	7.290e+05	0.000000	8.000000	166.000000	20.000000
75%	1.082e+06	0.000000	21.000000	178.000000	20.000000
max	1.304e+06	0.000000	146.000000	263.000000	20.000000

```
df.shape[0]
```

```
1095
```

```
df.shape[1]
```

```
20
```

```
df.columns
```

```
Index(['id', 'city', 'date', 'player_of_match', 'venue', 'neutral_venue',
       'team1', 'team2', 'toss_winner', 'toss_decision', 'winner',
       'result', 'result_margin', 'target_runs', 'target_overs',
       'super_over', 'method', 'umpire1', 'umpire2'],
      dtype=object)
```

```
df.dtypes
```

```
id                   int64
city                object
date                 object
player_of_match     object
venue                object
neutral_venue        int64
team1                object
team2                object
toss_winner          object
toss_decision        object
winner               object
result               object
result_margin       float
target_runs          int64
target_overs         int64
super_over           object
method               object
umpire1              object
umpire2              object
dtype: object
```

```
df.nunique()
```

```
id                 1095
city                 44
date                 1095
player_of_match     1090
venue                1095
neutral_venue        1
team1                1095
team2                1095
toss_winner          1095
toss_decision        1095
winner               1092
result               1095
result_margin       1076
target_runs          1095
target_overs         1095
super_over           1095
method               1095
umpire1              1095
umpire2              1095
dtype: int64
```

```
df.sample(5)
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
1	335983	Chandigarh	2008-04-19	MEK Hussey	Punjab Cricket Association Stadium, Mohali	0	Chennai Super Kings	Rajasthan Royals	Chennai Super Kings	bat	Chennai Super Kings	runs	33	241	20	N	NaN	MR Benson	SL Shah
2	335984	Delhi	2008-04-19	SC Ganguly	Feroz Shah Kotla	0	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	Delhi Daredevils	runs	9	238	20	N	NaN	Aleem Dar	GAF Cowie
3	335985	Mumbai	2008-04-20	MV Boucher	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Royal Challengers Bangalore	runs	5	234	20	N	NaN	SD Ranade	AV Jayaprakash
4	335986	Kolkata	2008-04-20	DJ Hussey	Eden Gardens	0	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	runs	5	222	20	N	NaN	BF Bowden	K Hariharan
5	335987	Jaipur	2008-04-21	SR Watson	Sawai Mansingh Stadium	0	Rajasthan Royals	Delhi Daredevils	Rajasthan Royals	bat	Rajasthan Royals	runs	55	200	20	N	NaN	Asad Rauf	RE Koertzen

```
df.sample(5)
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
1	335983	Chandigarh	2008-04-19	MEK Hussey	Punjab Cricket Association Stadium, Mohali	0	Chennai Super Kings	Rajasthan Royals	Chennai Super Kings	bat	Chennai Super Kings	runs	33	241	20	N	NaN	MR Benson	SL Shah
2	335984	Delhi	2008-04-19	SC Ganguly	Feroz Shah Kotla	0	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	Delhi Daredevils	runs	9	238	20	N	NaN	Aleem Dar	GAF Cowie
3	335985	Mumbai	2008-04-20	MV Boucher	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Royal Challengers Bangalore	runs	5	234	20	N	NaN	SD Ranade	AV Jayaprakash
4	335986	Kolkata	2008-04-20	DJ Hussey	Eden Gardens	0	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	runs	5	222	20	N	NaN	BF Bowden	K Hariharan
5	335987	Jaipur	2008-04-21	SR Watson	Sawai Mansingh Stadium	0	Rajasthan Royals	Delhi Daredevils	Rajasthan Royals	bat	Rajasthan Royals	runs	55	200	20	N	NaN	Asad Rauf	RE Koertzen

```
df.tail()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
1090	1304047	NaN	2019-05-05	AD Russell	Wankhede Stadium	0	Mumbai Indians	Kolkata Knight Riders	Mumbai Indians	field	Mumbai Indians	runs	34	232	20	N	NaN	Nitin Menon	CK Nandan
1091	1304048	NaN	2019-05-07	MS Dhoni	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Chennai Super Kings	Chennai Super Kings	bat	Royal Challengers Bangalore	runs	1	162	20	N	NaN	Nitin Menon	CK Nandan
1092	1304049	NaN	2019-05-08	KS Williamson	Eden Gardens	0	Kolkata Knight Riders	Sunrisers Hyderabad	Sunrisers Hyderabad	field	Sunrisers Hyderabad	runs	13	209	20	N	NaN	Nitin Menon	CK Nandan
1093	1304050	NaN	2019-05-10	AS Yadav	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Mumbai Indians	runs	14	188	20	N	NaN	Nitin Menon	CK Nandan
1094	1304051	NaN	2019-05-12	MS Dhoni	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Chennai Super Kings	Chennai Super Kings	bat	Chennai Super Kings	runs	6	71	20	N	NaN	Nitin Menon	CK Nandan

```
df.head()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
0	335982	Bangalore	2008-04-18	BB McCullum	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	field	Kolkata Knight Riders	runs	140	223	20	N	NaN	Asad Rauf	RE Koertzen
1	335983	Chandigarh	2008-04-19	MEK Hussey	Punjab Cricket Association Stadium, Mohali	0	Chennai Super Kings	Rajasthan Royals	Chennai Super Kings	bat	Chennai Super Kings	runs	33	241	20	N	NaN	MR Benson	SL Shah
2	335984	Delhi	2008-04-19	SC Ganguly	Feroz Shah Kotla	0	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	Delhi Daredevils	runs	9	238	20	N	NaN	Aleem Dar	GAF Cowie
3	335985	Mumbai	2008-04-20	MV Boucher	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Royal Challengers Bangalore	runs	5	234	20	N	NaN	SD Ranade	AV Jayaprakash
4	335986	Kolkata	2008-04-20	DJ Hussey	Eden Gardens	0	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	runs	5	222	20	N	NaN	BF Bowden	K Hariharan

```
df.tail()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
1090	1304047	NaN	2019-05-05	AD Russell	Wankhede Stadium	0	Mumbai Indians	Kolkata Knight Riders	Mumbai Indians	field	Mumbai Indians	runs	34	232	20	N	NaN	Nitin Menon	CK Nandan
1091	1304048	NaN	2019-05-07	MS Dhoni	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Chennai Super Kings	Chennai Super Kings	bat	Royal Challengers Bangalore	runs	1	162	20	N	NaN	Nitin Menon	CK Nandan
1092	1304049	NaN	2019-05-08	KS Williamson	Eden Gardens	0	Kolkata Knight Riders	Sunrisers Hyderabad	Sunrisers Hyderabad	field	Sunrisers Hyderabad	runs	13	209	20	N	NaN	Nitin Menon	CK Nandan
1093	1304050	NaN	2019-05-10	AS Yadav	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Mumbai Indians	runs	14	188	20	N	NaN	Nitin Menon	CK Nandan
1094	1304051	NaN	2019-05-12	MS Dhoni	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Chennai Super Kings	Chennai Super Kings	bat	Chennai Super Kings	runs	6	71	20	N	NaN	Nitin Menon	CK Nandan

```
df.drop_duplicates
```

```
<bound_method>
```

```
df.drop_duplicates()
```

	id	city	date	player_of_match	venue	neutral_venue	team1	team2	toss_winner	toss_decision	winner	result	result_margin	target_runs	target_overs	super_over	method	umpire1	umpire2
0	335982	Bangalore	2008-04-18	BB McCullum	M Chinnaswamy Stadium	0	Royal Challengers Bangalore	Kolkata Knight Riders	Royal Challengers Bangalore	field	Kolkata Knight Riders	runs	140	223	20	N	NaN	Asad Rauf	RE Koertzen
1	335983	Chandigarh	2008-04-19	MEK Hussey	Punjab Cricket Association Stadium, Mohali	0	Chennai Super Kings	Rajasthan Royals	Chennai Super Kings	bat	Chennai Super Kings	runs	33	241	20	N	NaN	MR Benson	SL Shah
2	335984	Delhi	2008-04-19	SC Ganguly	Feroz Shah Kotla	0	Delhi Daredevils	Kings XI Punjab	Delhi Daredevils	bat	Delhi Daredevils	runs	9	238	20	N	NaN	Aleem Dar	GAF Cowie
3	335985	Mumbai	2008-04-20	MV Boucher	Wankhede Stadium	0	Mumbai Indians	Royal Challengers Bangalore	Royal Challengers Bangalore	bat	Royal Challengers Bangalore	runs	5	234	20	N	NaN	SD Ranade	AV Jayaprakash
4	335986	Kolkata	2008-04-20	DJ Hussey	Eden Gardens	0	Kolkata Knight Riders	Deccan Chargers	Deccan Chargers	bat	Kolkata Knight Riders	runs	5	222	20	N	NaN	BF Bowden	K Hariharan
5	335987	Jaipur	2008-04-21	SR Watson	Sawai Mansingh Stadium	0	Rajasthan Royals	Delhi Daredevils	Rajasthan Royals	bat	Rajasthan Royals	runs	55	200	20	N	NaN	Asad Rauf	RE Koertzen

```
df.drop_duplicates(subset=['city'])
```